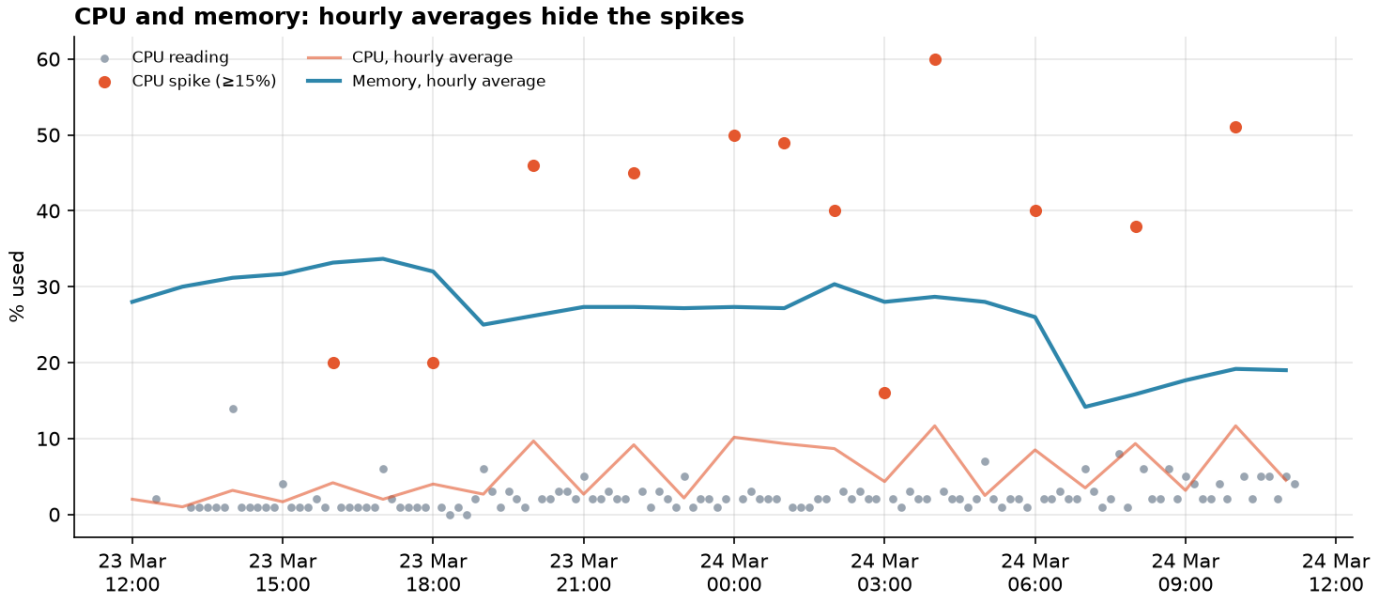
Server Health Analysis

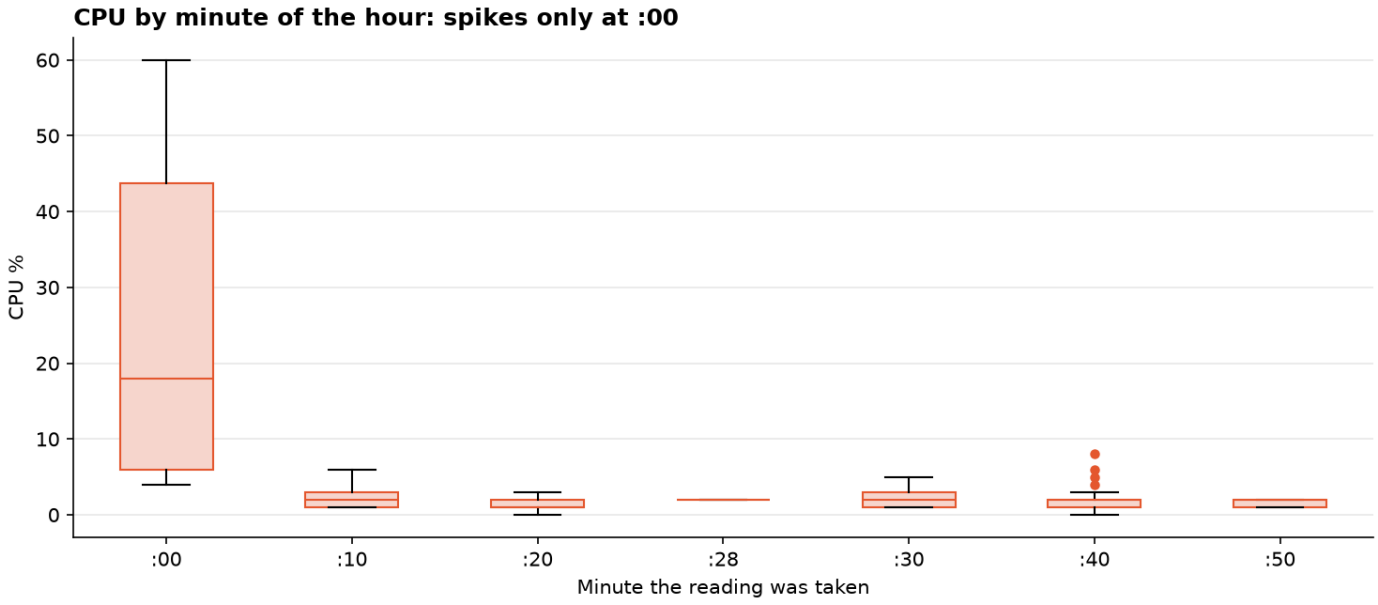
Scope

One server-monitoring log from 23 Mar 12:13 to 24 Mar 11:10 (24 hourly slots): resource readings every 10 minutes, security events (per-user process counts and per-IP connection counts) and system warnings. The raw text log was parsed into structured tables; every line parsed, none were dropped.

1. CPU load comes in short scheduled bursts

Average CPU is low (5.7%), but readings reach 60% (peak at 24 Mar 04:00). All 12 readings of 15% or more were taken at minute :00. Readings on the hour average 24.5%, every other reading 2.0%, and each burst lasts a single reading. That pattern points to a scheduled job (cron or similar). The spikes landed at 16:00, 18:00, 20:00, 22:00 (23 Mar) and 00:00, 01:00, 02:00, 03:00, 04:00, 06:00, 08:00, 10:00 (24 Mar). Hourly averaging flattens these bursts to at most 11.7%, so a dashboard built on hourly averages understates peak load about 5x.

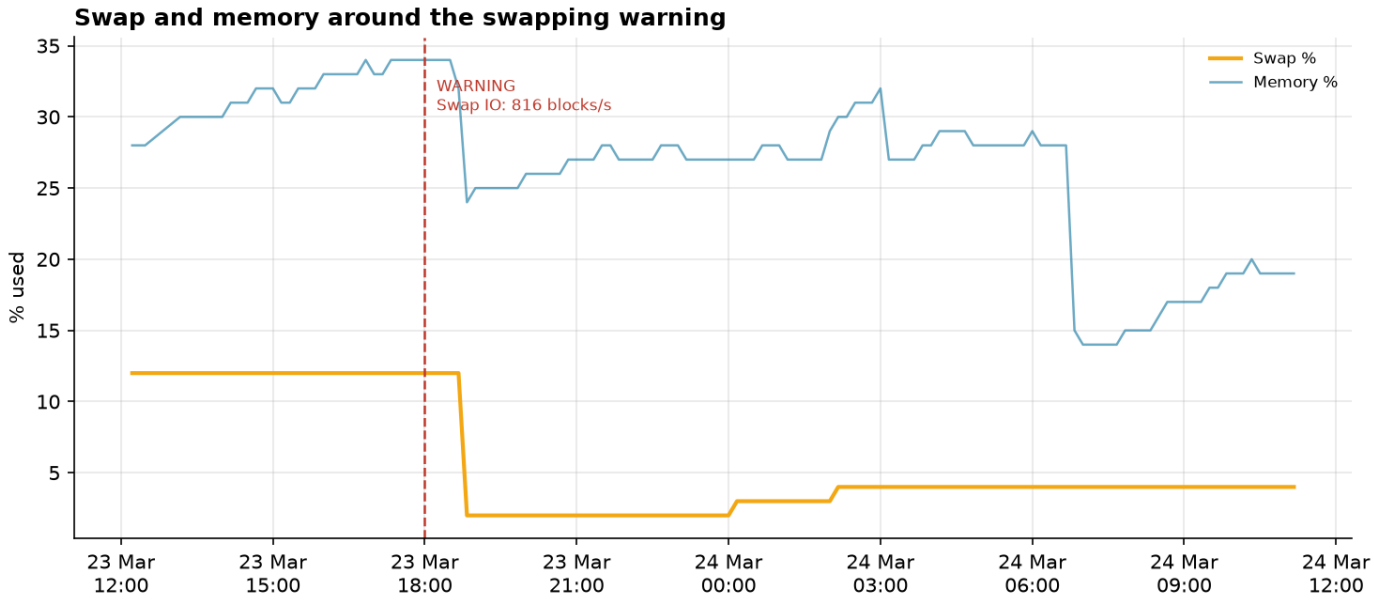




2. The swapping warning resolved itself

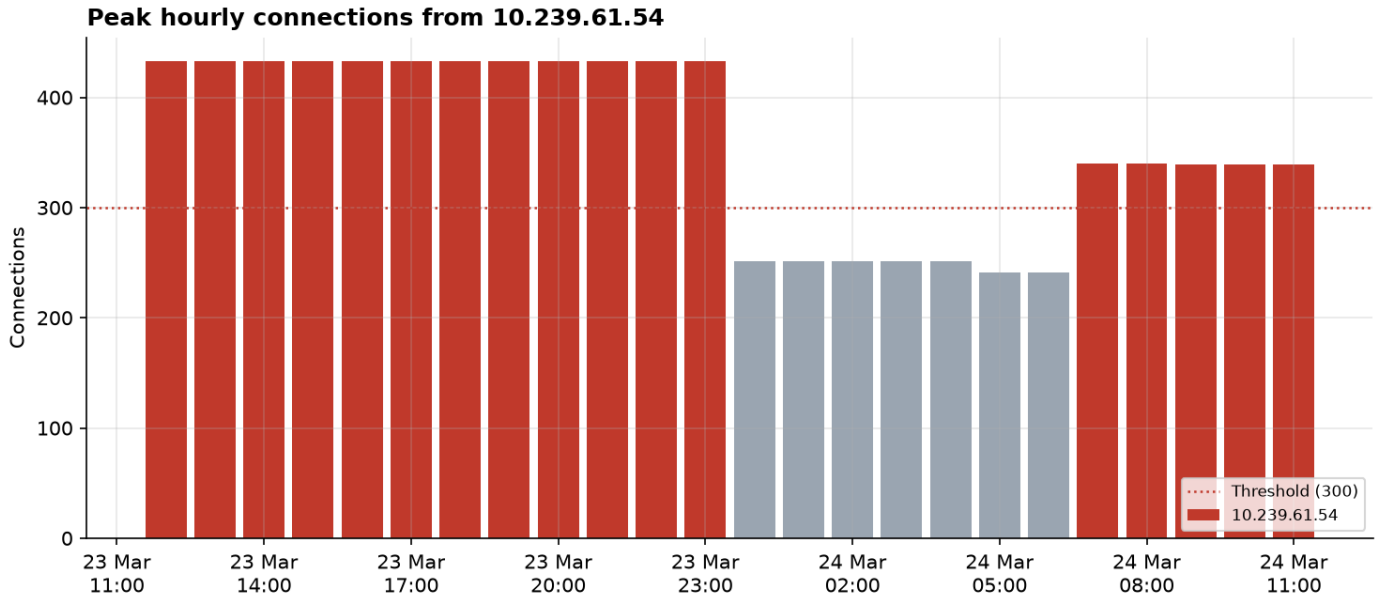
One warning was logged: 'Swap IO: 816 blocks/s' at 23 Mar 18:00. Swap sat at 12% before it. 50 minutes later swap

dropped to 2% and memory from 34% to 24% in the same reading, which is consistent with a process being stopped or restarted. Swap then crept back to 4% by the end of the log, worth watching over a longer window.



3. One internal IP holds hundreds of connections

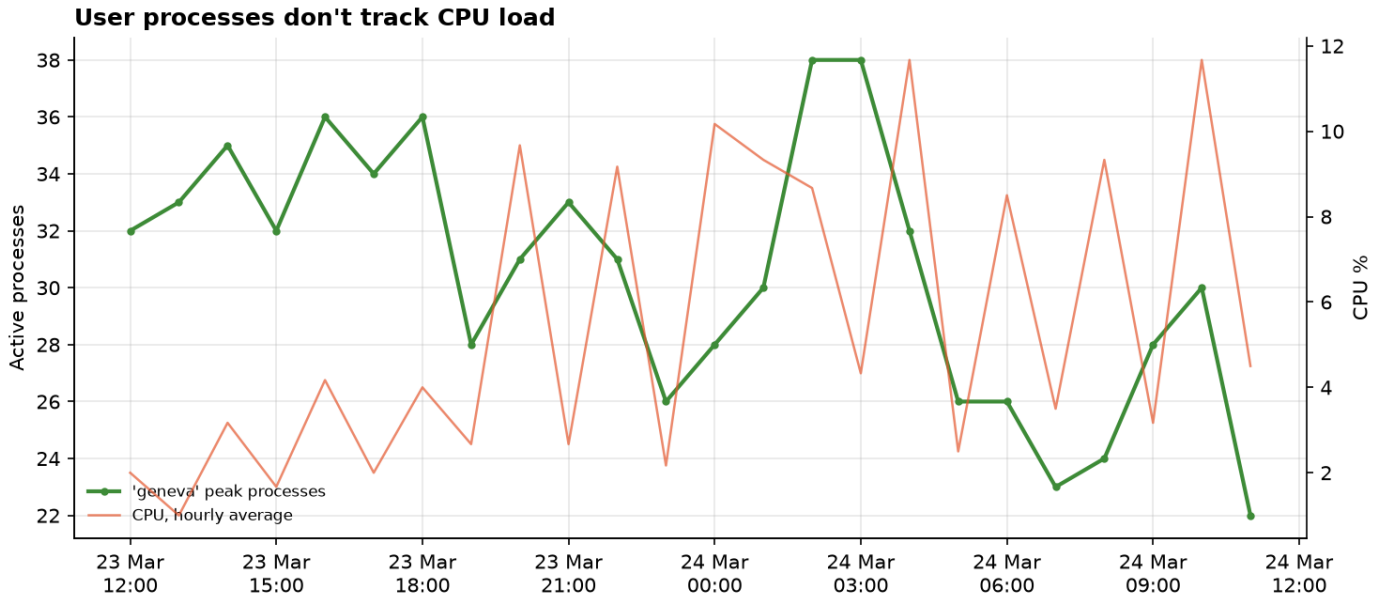
Every high-connection event came from 10.239.61.54, a private (internal) address. Its hourly peak was above 300 in 17 of 24 hours, up to 433. A single internal source at this volume is more likely a service (load balancer, proxy, connection pool) than an attack; the next step is to identify that host and set an expected baseline for it.



4. User processes don't explain the CPU load

User 'geneva' peaked at 38 active processes at 24 Mar 02:40, when CPU read 2%. At the CPU peak (60%, 04:00) the same user had 32. Across all hours the two are uncorrelated, whichever way CPU is summarised: r = -0.08 against the hourly average and r = 0.05 against the hourly peak. An early draft of this analysis blamed this user for the CPU load; the data doesn't support that. The bursts in finding 1 come from the scheduled job.

Server Health Analysis



Limits

This is a single day-long sample from one server, so it shows patterns, not trends. The pipeline and the Power BI model have date dimensions so more days of logs can be appended without changes.